TIC 41831052

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

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BLS EPHEMERIS
  Period        0.682620 d
  Epoch (t0)    3395.9339 BTJD
  Duration      0.5000 d
  Depth         0.028 %  (277 ppm)
  SNR           14.7

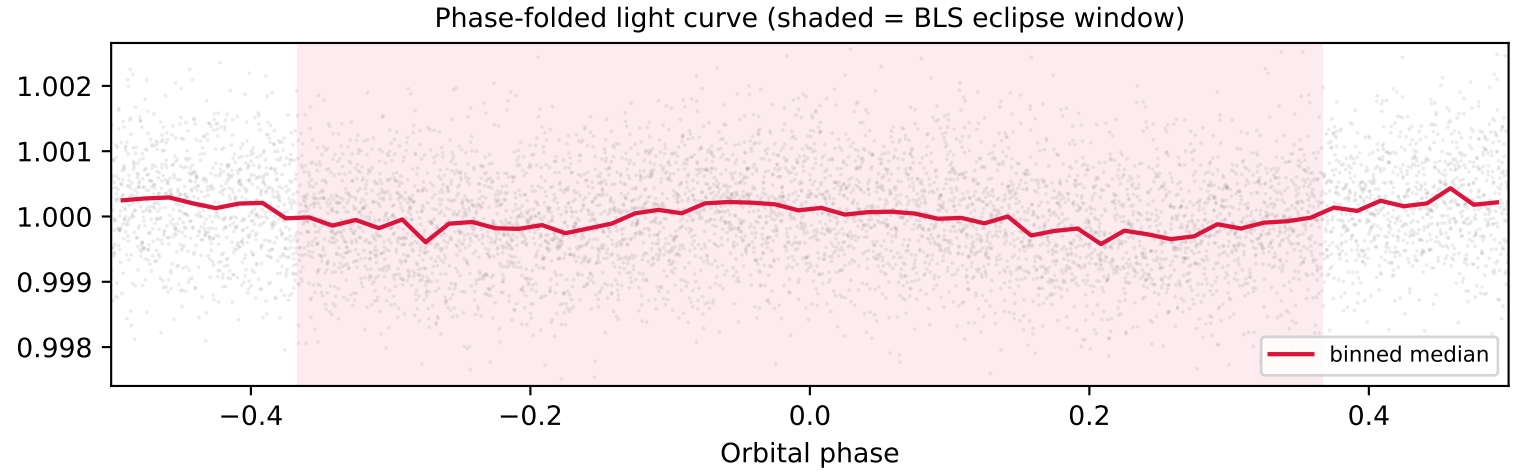
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      26  (gate >= 3)
  Robust depth ratio     1.01  (gate >= 0.50)
  Robust depth sigma     20.5  (gate >= 7.0)
  Odd/even z-score       0.21  (consistent)
  Secondary eclipse      not detected
```

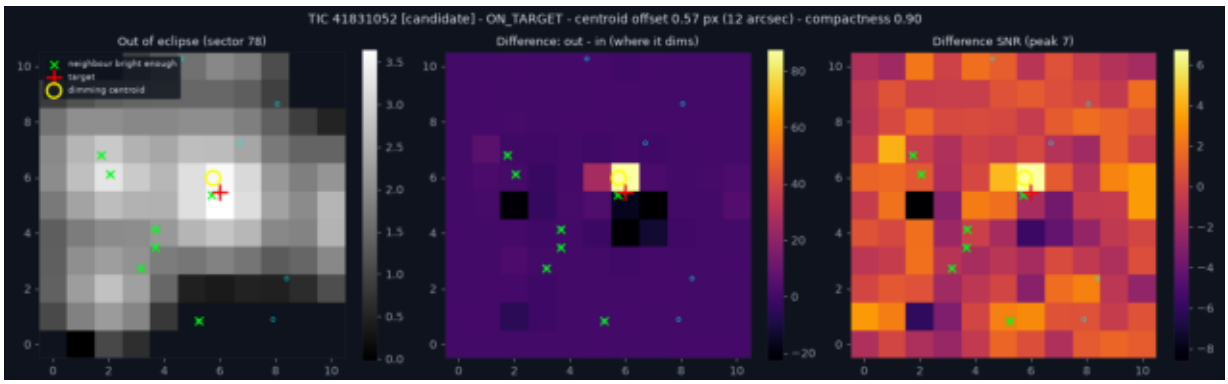
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CATALOG CROSS-MATCH (novelty)

  VSX                                  no match
  ASAS-SN                              no match
  Gaia eclipsing-binary table          no match
  Gaia variable classification         no match
  TESS EB catalog (Prsa+2022)          no match
  ExoFOP TOI list                      no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 78    light-curve dip in this sector: 12.6 sigma    t0 refined by 0.2200 phase
  Difference-image peak SNR   6.76   (gate >= 5.0)      Compactness   0.90   (gate >= 0.50)
  Dimming centroid offset     0.57 px (12 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 41831051
    0.60 px away, dTmag +5.53 - farther from the dimming than the target
```





Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.036 px (dimming centroid 0.566 px from target vs 0.602 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.